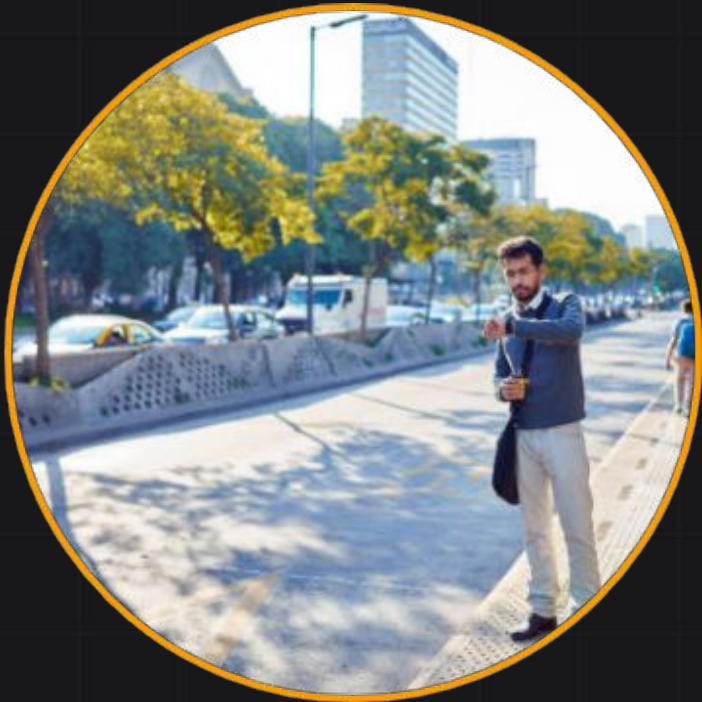


Predicting Busses

Real-Time Urban Mobility Analysis for
Málaga



The Urban Mobility Challenge



The Problem of Unpredictability

This project addresses a common challenge in urban mobility: the unpredictability of public transport.

Málaga buses can be early, late, or on time. This unpredictability is a practical issue in day-to-day life that this project aims to solve.

Project Goals

- 🎯 **Primary Goal:** Build an ML model for accurate, minute-by-minute predictions (on time, late, or early).
- 💡 **Key Insight:** Move beyond simple GPS tracking to provide predictive, actionable insights for Bus Line 31.

Key Deliverables

Historical Analysis

A historical analysis of Line 31's performance, using clustering to identify systemic delay patterns (e.g., specific times of day, locations) that cause a bus to be late.

Real-Time Engine

A real-time prediction engine built with Apache Spark. This engine will ingest live bus data, apply a trained regression model, and output a continuous stream of predictions.

Methodology

Clustering

Use clustering to analyze historical data and estimate if the bus line is late based on the time of day.



Regression

Use the dataset of real-time points and the static schedule to build a regression ML Model.



Prediction

The model will predict in minutes if the bus is late, on time, or is early, providing actionable insights.

Technology Stack

 **Framework:** Apache Spark

 **Language:** Python

 **Libraries:**

- SparkML (Regression Pipeline)
- Spark Structured Streaming
- Pandas (Data Exploration)



Data Sources

Emt Real-time API: Which provides live data for all active buses.

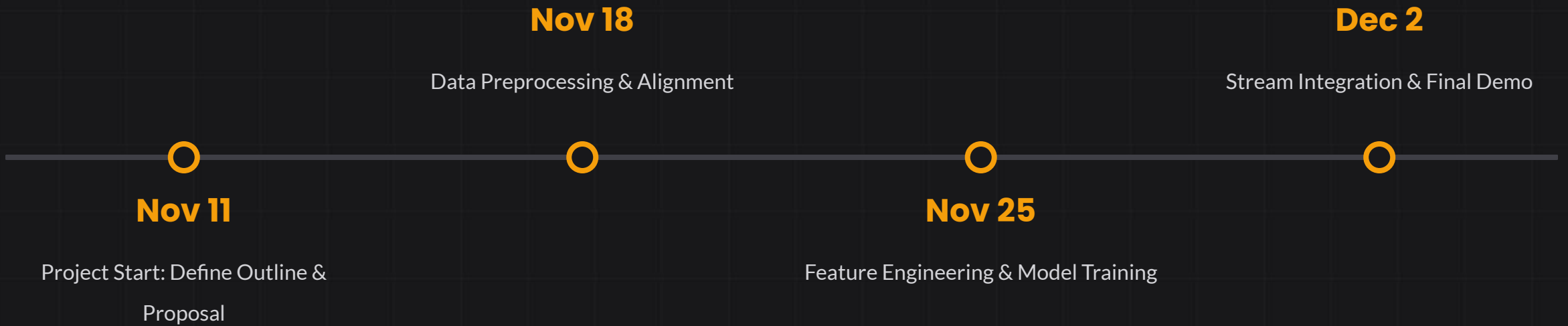
Emt Static GTFS: Which provides the official schedule including stop times.



Data Structure

Header	Meaning	Description
_id	Record ID	Internal unique identifier for the record
codBus	Bus Code	Unique Identifier for each physical bus
codLinea	Line Code	Identifier for the bus line or route
sentido	Direction	1 = outbound / 2 = return
lon. / lat.	Coordinates	Longitude & Latitude
codParIni	Starting/Stop Code	Initial Bus stop or current target bus stop
last_update	Last Update	Date and time of the most recent update

Project Timeline



Questions?

Thank you for your attention.

Image Sources



https://media.istockphoto.com/id/1701043022/photo/latina-american-commuter-looking-at-the-time-waiting-for-transportation.jpg?s=612x612&w=0&k=20&c=wU_ftttaMyEuqrUhsWUwGDR7hbXU9IhsoUBTs060jc=

Source: www.istockphoto.com



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